STATISTICS 5.1.1.

Representation of data

1. Types of data

Qualitative

(non - numerical)

- blood types: A, B, O, AB
- colors

Quantitative

(numerical)

discrete

e.g.: • # of letters in a word: 1, 2, 4, 8, 14, ...
• shoe size: $7\frac{1}{2}$, 9, 10, ...

continuous

any value in the range

e.g.: height (cm): 160.2, 160.24, ...

2. Stem - and - leaf diagrams

- small amounts of discrete data

- e.g.: Physics grade: 58, 55, 58, 61, 72, 79, 97, 67, 61, 77, 92, 64, 69, 62 and 53.

stem

- equal class widths
- ascending order

5	3 5 8 8
6	1 1 2 4 7 9
7	2 7 9
8	
9	2 7

Key: **5** | 3 represents a score of 53%

leaf:

- single digits only
- ascending order

EXAMPLE The heights of boys and girls in a year 11 class are given to the nearest cm in the back-to-back stem and leaf diagram below. Write out the data in full.

First boy, 8|16|, has height 168 cm. The boys are read backwards.

Key 8|16|5 means
Boys 168 cm and girls 165 cm

Boys		Girls
	15	9
8	16	1, 5, 9
9, 8, 1	17	0, 2, 3, 5
5, 2	18	0
1	19	

First girl, |15|9, has height 159 cm.

Boys: 168, 171, 178, 179, 182, 185, 191

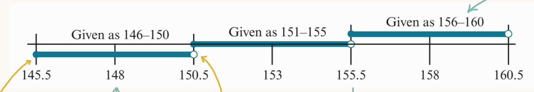
Girls: 159, 161, 165, 169, 170, 172, 173, 175, 180

3. Histograms

- continuous data * discrete data too
  - numbers are rounded up / down to fit into classes

class

e.g.:



lower boundary

upper boundary

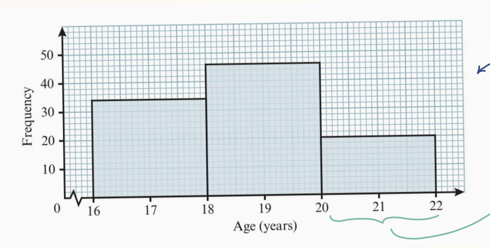
class mid-value (midpoint)

class width

Ex 1.

Age (*A* years)	$16 \leqslant A < 18$	$18 \leqslant A < 20$	$20 \leqslant A < 22$
No. students (*f*)	34	46	20

grouped frequency table



frequency diagram
- no gaps between columns

= class width

"proportional to"

	First	Second	Third
Area	2 × 34 = 68	2 × 46 = 92	2 × 20 = 40
Frequency	34	46	20

$\boxed{\text{area} \propto \text{frequency}}$

Ex. 2

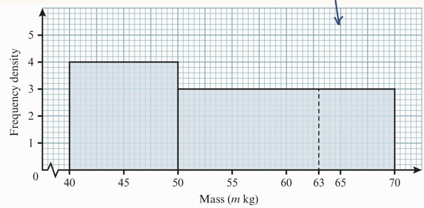
Mass (*m* kg)	$40 \leqslant m < 50$	$50 \leqslant m < 70$
No. children (*f*)	40	60

Mass (*m* kg)	$40 \leqslant m < 50$	$50 \leqslant m < 70$
No. children (*f*)	40	60
Class width (kg)	10	20
Frequency density	40 ÷ 10 = 4	60 ÷ 20 = 3

histogram

$$\boxed{\text{frequency density} = \frac{\text{frequency}}{\text{class width}}}$$

= frequency per standard interval



$$\boxed{\text{frequency} = \text{f.density} \times \text{class width}}$$

* assume values are evenly spread in whole class interval

4. Cumulative frequency graphs

· continuous data

Ex 1.

boundaries:

$0.5 \leq l < 2.5$

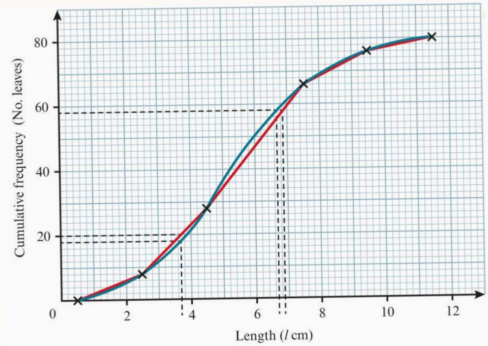
Lengths (cm)	1–2	3–4	5–7	8–9	10–11
No. leaves (f)	8	20	38	10	4

Lengths (l cm)	Addition of frequencies	No. leaves (cf)
$l < 0.5$	0	0
$l < 2.5$	$0+8$	8
$l < 4.5$	$0+8+20$	28
$l < 7.5$	$0+8+20+38$	66
$l < 9.5$	$0+8+20+38+10$	76
$l < 11.5$	$0+8+20+38+10+4$	80

upper boundary

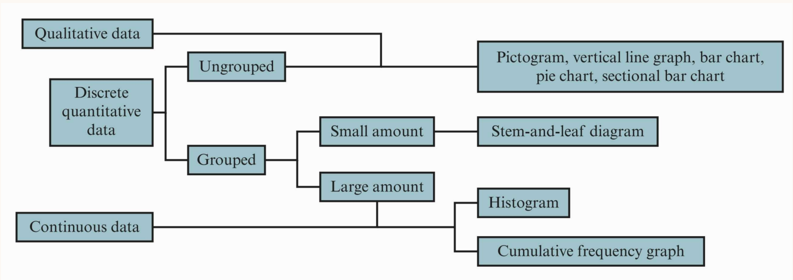
cf = cumulative frequency

= Σ frequency of all values up to a certain point



* draw a curved line to connect the dots

5. Comparing different data representations



Measures of Central Tendency

1. Three types of averages

EXAMPLE Find the mean, median and mode of the following list of data: 2, 3, 6, 2, 5, 9, 3, 8, 7, 2

Put in order first: 2, 2, 2, 3, 3, 5, 6, 7, 8, 9

Mode = **2**

Mean = $\frac{2+2+2+3+3+5+6+7+8+9}{10} = \underline{\mathbf{4.7}}$

Median = average of 5th and 6th values = **4**

mode : most frequently occurring data value

mean = $\bar{x} = \frac{\Sigma x}{n} = \frac{\Sigma fx}{\Sigma f}$

$\frac{\text{sum of values}}{\text{number of values}}$

sum ("sigma")

frequency

median = $\left(\frac{n+1}{2}\right)$th value

the middle value when all values are placed in order of size

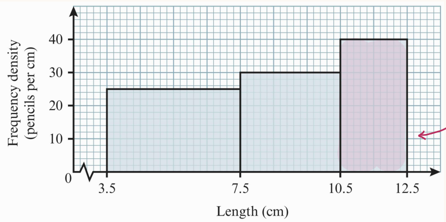
2. Mode and modal class

Ex : lengths of 270 pencils

Length (x cm)	No. pencils (f)
4–7	100
8–10	90
11–12	80

→

Length (x cm)	No. pencils (f)	Width (cm)	Frequency density
$3.5 \leqslant x < 7.5$	100	4	$100 \div 4 = 25$
$7.5 \leqslant x < 10.5$	90	3	$90 \div 3 = 30$
$10.5 \leqslant x < 12.5$	80	2	$80 \div 2 = 40$



modal class : class with highest frequency density

3. Means from grouped frequency tables

→ you'll have to estimate using mid-class values

Height of tree to nearest m	Mid-class value x	Number of trees f	fx
0 - 5	2.75	26 (26)	71.5
6 - 10	8	17 (43)	136
11 - 15	13	11	143
16 - 20	18	6	108
	Totals	60 (= Σf)	458.5(= Σfx)

Lower class boundary = 0.
Upper class boundary = 5.5.
So the mid-class value = (0 + 5.5) ÷ 2 = 2.75.

Estimated mean = $\frac{458.5}{60}$ = **7.64 m**

* coded data : transform values (adding a +ve / -ve constant)
↳ easier to handle

Ex : Find mean of 101 , 103 , 104 , 109 , 113

$\downarrow -100$ put it back!

1 , 3 , 4 , 9 , 13

↳ $\bar{x} = \dfrac{1+3+4+9+13}{5} + 100 = 6 + 100 = 106$

For ungrouped data , $\boxed{\bar{x} = \dfrac{\Sigma(x-b)}{n} + b}$

For grouped data , $\boxed{\bar{x} = \dfrac{\Sigma(x-b)f}{\Sigma f} + b}$

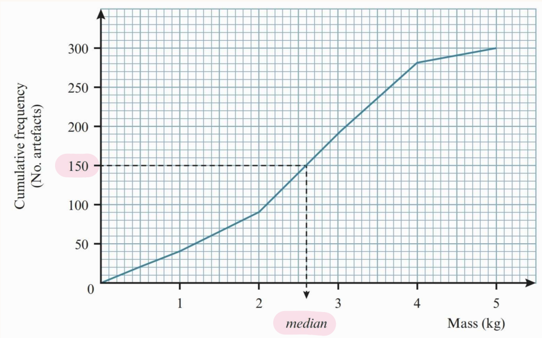
$\bar{x} = \text{mean}(x-b) + b$

4. Median for grouped data

→ we estimate , yet again , with a graph

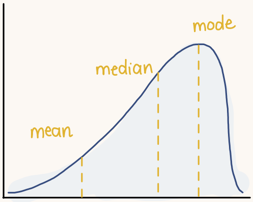
E.x.: masses of 300 museum artefacts

for an estimate,
median = $\dfrac{n}{2}$ th value



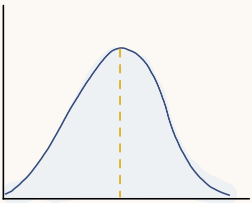
* skewness (which average is heavily affected by outliers?)

negative skew



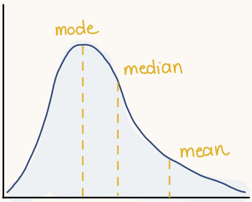
mean < median < mode

symmetrical



mean = median = mode

positive skew



mode < median < mean

Measures of Variation

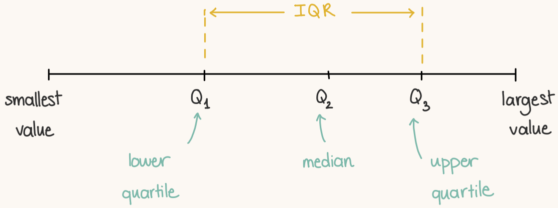
↳ to measure: how widely spread out the values are. / dispersion

1. Range

$\boxed{\text{range} = \text{highest value} - \text{lowest value}}$

* heavily affected by extremes / outliers

2. Interquartile range (IQR) + percentiles



$\frac{1}{4}n$ or $\frac{1}{4}\Sigma f$

$\frac{3}{4}n$ or $\frac{3}{4}\Sigma f$

$\boxed{IQR = Q_3 - Q_1}$

* relatively unaffected by outliers

* for grouped data
total frequency $n = \Sigma f$

* for ungrouped data

- even # of values

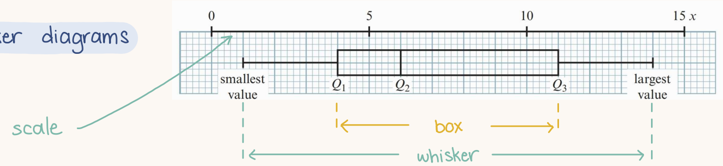
1st	2nd	3rd	4th	5th	6th	7th	8th
2	5	9	13	29	33	49	55

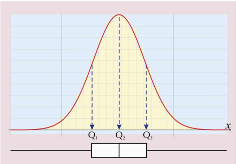
Q_1 (between 2nd and 3rd), Q_2 (between 4th and 5th), Q_3 (between 6th and 7th)

- odd # of values

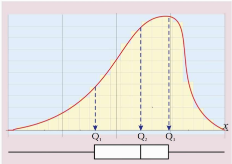
1st	2nd	3rd	4th	5th	6th	7th
6	17	39	43	69	73	77
	Q_1		Q_2		Q_3	

* box - and - whisker diagrams

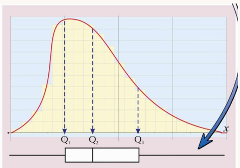




If $Q_3 - Q_2 = Q_2 - Q_1$ then the skewness is zero.



If $Q_3 - Q_2 < Q_2 - Q_1$ then the distribution is negatively skewed.



If $Q_3 - Q_2 > Q_2 - Q_1$ then the distribution is positively skewed.

Var SD

3. Variance and standard deviation

↳ measures: how close a value is to the mean

standard deviation = $\sqrt{\text{variance}}$

- for ungrouped data:

$\bar{x} = \dfrac{\sum x}{n}$;

$$\boxed{s = \sqrt{\frac{\sum (x-\bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}}$$

- for grouped data:

$\bar{x} = \dfrac{\sum fx}{\sum f}$;

$$\boxed{s = \sqrt{\frac{\sum (x-\bar{x})^2 f}{\sum f}} = \sqrt{\frac{\sum x^2 f}{\sum f} - \bar{x}^2}}$$

EXAMPLE Find the mean and standard deviation of the following numbers: 2, 3, 4, 4, 6, 11, 12

1) Find the total of the numbers first: $\sum x = 2 + 3 + 4 + 4 + 6 + 11 + 12 = 42$

2) Then the mean is easy: Mean $= \bar{x} = \frac{\sum x}{n} = \frac{42}{7} = 6$

3) Next find the sum of the squares: $\sum x^2 = 4 + 9 + 16 + 16 + 36 + 121 + 144 = 346$

4) Use this to find the variance: Variance, $s^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{346}{7} - 6^2 = 49.43 - 36 = 13.43$

5) And take the square root to find the standard deviation: Standard deviation $= \sqrt{13.43} = 3.66$ to 3 sig. fig.

combined sets of data

mean of x and y combined $= \dfrac{\sum x + \sum y}{n_x + n_y}$

variance of x and y combined $= \boxed{\dfrac{\sum x^2 + \sum y^2}{n_x + n_y} - \left(\dfrac{\sum x + \sum y}{n_x + n_y}\right)^2}$